

Quantifying the Relationship Between Digital Behavior and Mental Health Outcomes in Adolescents

Behavioral research brief · Quantitative Social Science (2025)

Data provenance: the analysis below runs on a **simulated** survey dataset ($n = 1,200$) built from a documented data-generating process, so the pipeline is fully reproducible without restricted human-subjects data. The identical code (`run_all.py`) produces this brief verbatim when pointed at a real adolescent survey with the same fields. Effect sizes are therefore illustrative of the method, not empirical findings about real teenagers.

Problem

Adolescent screen time has risen sharply, and clinicians and educators increasingly ask whether heavier digital engagement tracks with worse sleep, higher stress, and weaker academic performance. The policy-relevant question is not only *whether* social-media use correlates with distress, but *through what pathway* — because the pathway determines where intervention is cheapest. This brief quantifies the associations, isolates the role of sleep as a mechanism, and packages exposure into a single screening index usable at the student level.

Data

The dataset contains 1,200 adolescents (ages 13–18) with self-reported daily social-media hours, screen time before sleep, sleep duration, weekly physical activity, an in-person social-interaction score (1–10), primary platform, plus two outcomes: a stress index (0–10) and academic performance (GPA, 0–4). Roughly 3–4% of behavioral fields were missing and imputed (median for numeric, mode for categorical); scale variables were z-standardized for comparability. Social-media use was split into Low / Medium / High tertiles for group contrasts.

Methods

Analysis proceeds in five steps. (1) **Descriptive contrasts** of each outcome across usage tertiles. (2) A **Pearson correlation matrix** over all continuous measures. (3) **Welch two-sample t-tests** (with Cohen's d and 95% confidence intervals) comparing high- versus low-usage students and TikTok versus Instagram users, plus one-way ANOVA across tertiles. (4) **OLS regression** of each outcome on the five behaviors and demographic controls (age, gender, platform), reporting unstandardized and standardized coefficients, variance-inflation factors, and residual diagnostics. (5) A **mediation analysis** decomposing the social-media $\rightarrow$ stress association into a direct path and an indirect path through sleep, with a 5,000-draw nonparametric bootstrap confidence interval for the indirect effect. Regression inference is computed in closed form (validated against scikit-learn to 1×10^{-12}). Finally, a composite **Digital Exposure Risk Index (DERI)** is constructed and correlated with both outcomes.

Findings

Heavier users sleep less, report more stress, and earn lower grades — and the gradient is monotonic. Moving from the lowest to highest usage tertile, mean sleep falls from 8.3 to 7.0 hours, stress rises from 3.8 to 6.0, and GPA drops from 3.63 to 2.96. The high-versus-low contrasts are large and unambiguous:

Contrast (high – low usage)	Difference	95% CI	Cohen's d	p
Stress (0–10)	+1.62	[1.46, 1.78]	1.15	< .001

Sleep (hrs)	-0.90	[-0.98, -0.81]	-1.23	< .001
GPA	-0.49	[-0.54, -0.43]	-1.04	< .001

Correlations point the same way: social-media hours correlate **+0.61** with stress and **-0.65** with sleep, while sleep correlates **-0.58** with stress and **+0.58** with GPA. A one-way ANOVA across tertiles confirms the stress gradient ($F = 260$, $p < .001$). TikTok users report modestly more daily use and higher stress than Instagram users ($d = 0.35$ and 0.24 respectively), a small but reliable platform difference.

The associations survive adjustment. In the regression on stress ($R^2 = 0.49$), each additional daily hour of social media predicts **+0.42** stress points ($\beta = 0.37$) and each additional hour of sleep predicts **-0.53** ($\beta = -0.28$); physical activity and in-person interaction are independently protective. The academic-performance model ($R^2 = 0.49$) mirrors this — social-media use and low sleep both depress GPA. Screen time *immediately* before sleep is not significant once sleep duration is in the model, indicating its effect operates through total sleep rather than independently. All variance-inflation factors are below 2.1, so multicollinearity is not distorting the estimates.

About a third of social media's link to stress runs through sleep. Decomposing the total social-media $\rightarrow$ stress effect ($c = +0.71$): more use predicts less sleep ($a = -0.40$), less sleep predicts more stress ($b = -0.59$), yielding an **indirect effect of +0.235** (bootstrap 95% CI [0.19, 0.28]; Sobel $z = 10.1$). A direct effect remains ($c' = +0.47$), so sleep is a **partial mediator accounting for ~33% of the association** — the rest operates through other channels. This is the actionable result: protecting sleep recovers a meaningful share of the stress burden without eliminating screen use entirely.

Sleep as a mediator of social media $\rightarrow$ stress

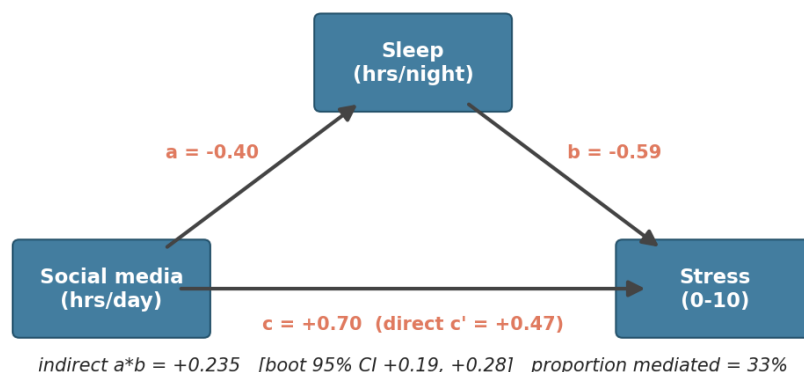


Figure 1. Sleep partially mediates the social-media $\rightarrow$ stress association (indirect effect +0.235; ~33% of the total effect).

A simple composite index screens effectively. The DERI combines (standardized) social-media hours, pre-sleep screen time, and inverted sleep duration with equal weights. It correlates **+0.62** with stress and **-0.61** with GPA. A PCA-derived weighting reproduces it almost exactly (first component explains 75% of shared variance; correlation with the equal-weight index ≈ 1.00), so the transparent equal-weight version is justified. Sorting students into DERI quartiles produces a clean dose-response: mean stress climbs from 3.6 (lowest-risk quartile) to 6.2 (highest), and GPA falls from 3.69 to 2.89.

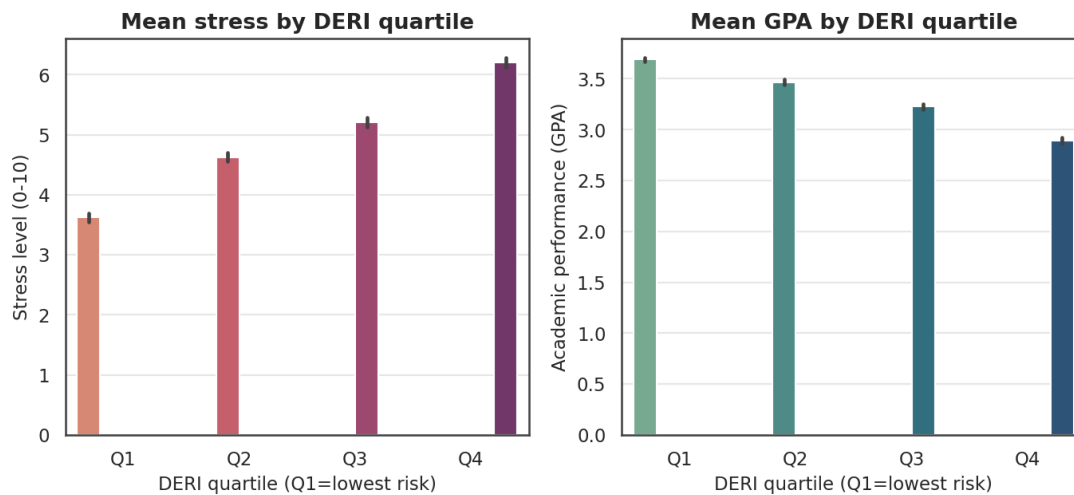


Figure 2. Stress rises and GPA falls monotonically across Digital Exposure Risk Index quartiles (Q1 = lowest risk).

Implications

Three takeaways follow. First, **sleep is the leverage point**: because a third of the stress association is mediated by sleep loss, sleep-protective measures (curfews on pre-bed device use, later school start times, sleep-hygiene education) plausibly capture much of the benefit of broader screen-time limits at lower cost. Second, **a lightweight index is enough to triage**: the DERI's strong, monotonic relationship with both outcomes means schools could flag higher-risk students from three routinely collected behaviors rather than fielding lengthy instruments. Third, **protective behaviors matter independently**: physical activity and in-person social contact predict lower stress and higher GPA even after adjusting for screen use, suggesting that *substituting* offline activity may be more effective messaging than *subtracting* screen time alone.

Limitations

The design is cross-sectional, so the regression and mediation estimates describe associations and a modeled pathway, not established causation; reverse causality (stress driving both poor sleep and compensatory scrolling) is plausible and untestable here. Self-report introduces measurement error, and — as noted above — these specific magnitudes come from simulated data and should be replaced with estimates from a real survey before any findings are reported.

Reproducibility: python run_all.py regenerates every figure, table, and the cleaned, DERI-scored dataset. Regression inference is implemented from first principles in ols.py.